

What is a Javascript ?

JavaScript is a programming language used to make websites and applications interactive.

Think of a website as:

- HTML → Structure
- CSS → Appearance
- JavaScript → Behavior / Logic

Example

```
alert("Welcome to my website!");
```

JavaScript tells the computer:

"When this code runs, show a welcome message."

Real-world examples

JavaScript can be used for:

- Button clicks
- Form validation
- Calculations
- Shopping carts
- Games
- Animations
- Login systems
- Web applications

Why do we need Javascript ?

Without JavaScript, a webpage is mostly static.

Example: Shopping Website

User clicks:

"Add to Cart"

JavaScript can:

1. Detect the click
2. Add the product to the cart
3. Increase cart count
4. Calculate the total price
5. Update the screen
- 6.

Simple idea

USER ACTION



JAVASCRIPT



LOGIC



CHANGE THE PAGE

JavaScript connects user actions with program behavior.

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Where Does Javascript Run ?

JavaScript can run in different environments.

1. Browser

Examples:

- Chrome
- Edge
- Firefox
- Safari

Used for:

Websites

Web Applications

Browser Games

2. Server

With Node.js, JavaScript can run on a server.

Used for:

APIs

Backend Applications

Database Operations

Authentication

Simple picture

Browser



JavaScript



Server



JavaScript / Node.js

So JavaScript is not limited to just the browser.

How do we write javascript ?

JavaScript instructions are written as code.

Example:

```
let name = "John";
```

```
console.log(name);
```

Here:

- `let` → creates a variable
- `name` → variable name
- `"John"` → value
- `console.log()` → displays the value in the console

Think of it like this:

Variable

↓

name

↓

"John"

A variable is like a box that stores information.

Javascript Data

Programs need to work with different kinds of information.

Common data types

String

```
let name = "John";
```

Number

```
let age = 25;
```

Boolean

```
let isPlaying = true;
```

Only:

true

False

Example

```
let playerName = "Alex";
```

```
let score = 100;
```

```
let isAlive = true;
```

JavaScript makes Decision

Programs often need to make decisions.

Example

A game checks:

Is the player's health greater than 0?

```
if (health > 0) {  
  console.log("Player is alive");  
}
```

If the condition is true:

Health > 0



YES



Player is alive

If it is false:

Health ≤ 0



NO



Player is dead

Real-world examples

JavaScript can make decisions such as:

- Is the password correct?
- Is the player alive?
- Is the score high enough?
- Is the cart empty?
- Is the user logged in?